Data Science project: Database of past climates data

ICE RAFTED DEBRIS

The aim of this project is to compare model simulations of sea ice extension in different climate states with one of the indicators of past sea ice extension from geological archives, i.e. the Ice Rafter Debris (IRD). Other people started working on this project, so the task is to review, correct (where necessary) and extend their work.

Deliverable: Brief relation/readme doc, organized data files, notebooks, yml file.

Work should be carried out in a new, dedicated conda environment, that needs to be stored and delivered in the form of a yml file: https://docs.conda.io/projects/conda/en/4.6.0/_downloads/52a95608c49671267e40c689e0bc00ca/conda-cheatsheet.pdf

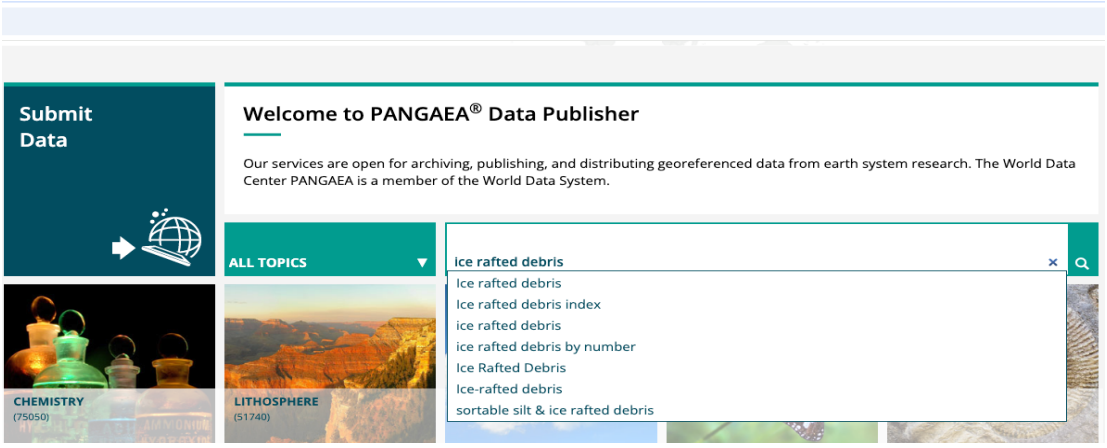
2. Paleoclimate proxy data

Because we do not have direct observation of sea ice extent prior to the satellite or in any case the observational era, we need to resort to geological archives to infer the past extension of sea ice, based paleoclimate proxy data.

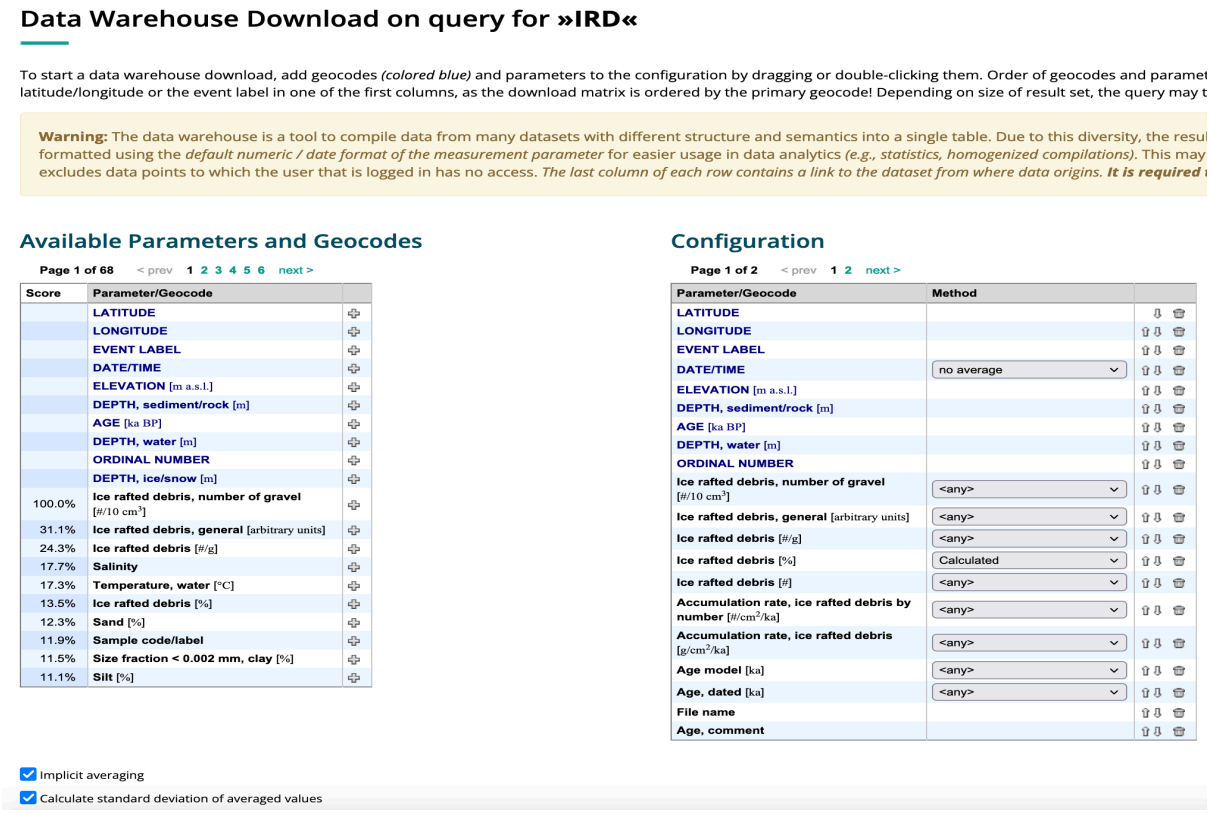
In this case we will focus on the "Ice Rafted Debris" (IRD) as a proxy for past sea ice extension.

2.1 Search & Download

There are several data repositories that could be consulted. Here we start from https://pangaea.de/. Include in the download all the fields that appear when searching for "ice rafted debris" and "IRD".



There are some tools available in Pangaea for downloading multiple datasets together, i.e. via "warehouse". It is possible to determine which dimensions/metadata to export as well as the main variables. Here is an example of selected fields to download from a list of possibilities.



Data Warehouse Download on query for »IRD«

To start a data warehouse download, add geocodes *(colored blue)* and parameters to the configuration by dragging or double-clicking them. Order of geocodes and paramet
latitude/longitude or the event label in one of the first columns, as the download matrix is ordered by the primary geocode! Depending on size of result set, the query may t

Warning: The data warehouse is a tool to compile data from many datasets with different structure and semantics into a single table. Due to this diversity, the resul
formatted using the *default numeric / date format of the measurement parameter* for easier usage in data analytics *(e.g., statistics, homogenized compilations)*. This may
excludes data points to which the user that is logged in has no access. *The last column of each row contains a link to the dataset from where data origins.* ***It is required***

Available Parameters and Geocodes

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Score	Parameter/Geocode
	LATITUDE
	LONGITUDE
	EVENT LABEL
	DATE/TIME
	ELEVATION [m a.s.l.]
	DEPTH, sediment/rock [m]
	AGE [ka BP]
	DEPTH, water [m]
	ORDINAL NUMBER
	DEPTH, ice/snow [m]
100.0%	Ice rafted debris, number of gravel [#/10 cm^3]
31.1%	Ice rafted debris, general [arbitrary units]
24.3%	Ice rafted debris [#/g]
17.7%	Salinity
17.3%	Temperature, water [°C]
13.5%	Ice rafted debris [%]
12.3%	Sand [%]
11.9%	Sample code/label
11.5%	Size fraction < 0.002 mm, clay [%]
11.1%	Silt [%]

Configuration

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Parameter/Geocode	Method
LATITUDE	
LONGITUDE	
EVENT LABEL	
DATE/TIME	no average
ELEVATION [m a.s.l.]	
DEPTH, sediment/rock [m]	
AGE [ka BP]	
DEPTH, water [m]	
ORDINAL NUMBER	
Ice rafted debris, number of gravel [#/10 cm^3]	<any>
Ice rafted debris, general [arbitrary units]	<any>
Ice rafted debris [#/g]	<any>
Ice rafted debris [%]	Calculated
Ice rafted debris [#]	<any>
Accumulation rate, ice rafted debris by number [#/cm^2/ka]	<any>
Accumulation rate, ice rafted debris [g/cm^2/ka]	<any>
Age model [ka]	<any>
Age, dated [ka]	<any>
File name	
Age, comment	

Implicit averaging

Calculate standard deviation of averaged values

2.2 Explore and organize

In the example notebook (IRD_test_01_bis.ipynb) some preliminary data exploration steps are carried out, as well as some data quality screening. These operations could include checking for sites “badly” located on land (whereas all should be from the ocean; sometimes latitude and longitude values may have been inverted), as well as checking for negative values for variables that are zero or positive by definition.

Then some details must be recovered from the different types of variables included, in order to understand them.

Some of the sites will include multiple values (from the same author/study) listed as separate entries; it would be useful to concatenate those as time/depth series. Defining ad hoc classes of objects to organize the dataset is one option, minimally explored in (ird_classes.ipynb).

At the end we would like to extract subsets of the data relevant for specific time slices.

A notebook is available: IRD.ipynb

STATUS: most operations done, but not all, and there are still problems.

1. Enhance the info on maps for the categories we have, highlighting the type of indicator, whether for a site we have a single point or a depth/time series, whether a site is “suspicious” (see point below).
2. The maps with site location shows data in locations in which we do not expect to have IRD. Generally, we do not expect IRD at low latitudes (< |30| deg.), so those are suspicious sites, and sites falling on land are not okay of course.
3. A working hypothesis of previous students was to review the coordinates correction. We have a few potential problems: Lat/Lon inverted; Lon can be expressed in different ways (range -180 to +180 degrees East, range 180 W to 180 E, range 0 to 360 E). I now suspect this is not the problem (not the main one, in any case). Let’s exclude this “correction” for now.

4. Perform a few manual checks on sites left with obvious problems. Some sites may be associated with the same publication, which can help solve multiple problems simultaneously, sometimes.
5. Use additional data repositories to expand the dataset (see all the search keywords above): https://www.ncei.noaa.gov/access/paleo-search/ and https://zenodo.org/ .

Here are a few comments to consider, concerning point 2.

1. Un problema che posso rilevare è che sono stati inclusi siti non pertinenti.
Ho aperto su pangea i primi elemento casuale per il controllo manuale (low lat)

	tipo_completo	Event Label	Origin of Values	lat_orig	lon_orig	Depth sed [m]
0	low_latitude_depth_0	EN066-24PG	https://doi.pangaea.de/10.1594/PANGAEA.227904	3.847000	-20.367000	0.00
1	low_latitude_depth_gt0	KNR166-2-51	https://doi.pangaea.de/10.1594/PANGAEA.824555	24.411670	-83.218900	0.05

In effetti le coordinate sono corrette, ma nel dataset originale non ci sono colonne rilevanti per "IRD" - in effetti in "ice_rafted_debris.tab" le colonne *ird* sono vuote (nel primo; non nel secondo).

2. Un'altra casistica è che abbiamo in effetti recuperato siti non-marini:

3	on_land_depth_gt0	Lake_Sidi_Ali_2012	https://doi.pangaea.de/10.1594/PANGAEA.960342	33.068323	-5.000662	64.39

3. Lo shapefile terre emerse è troppo impreciso, forse?

2	on_land_depth_0	PS2642-2	https://doi.pangaea.de/10.1594/PANGAEA.54277	72.790000	-25.821667	0.00